

Your first thirty minutes

From a finished install to a working account, a file and a second user signing in over ssh.

This page assumes SD Core is installed and you have never used this port before. It is a walkthrough, not a reference — every step links to the page that explains it properly.

Do this first, before anything else.

1. Sign out of Windows and back in

Not optional, and not a tidiness step. The installer put your Windows account into the `sdusers` group, and **Windows fixes group membership when you sign in**. Until you get a new logon token you cannot read the database at all.

The symptom looks like a broken install, not a permissions problem, which is why it is step 1. The post-install check knows this: the run it offers at the end of the installer is **always the incomplete one**, for exactly this reason.

Sign out, sign back in, then run **Check the SD installation** from the Start Menu. It closes on a keypress.

2. Start SD

```
sd
```

You land in **the SD account named after your Windows login**. Nothing asks for a password — Windows has already authenticated you.

SD is already running. It is a Windows service, `String Database (SD)`, and Windows starts it at every boot. You do not type `sd -start`. See Running SD.

If `sd` answers *Account ... not in register*, you are in the wrong account or step 1 has not taken effect. If it answers *not registered for SD use*, you are not in `sdusers`.

3. Look around

```
who
listf
term
```

`who` the account you are standing in

`listf` the files in it

`term` your terminal type and page size — should say `Device : windows`

If `term` says something else and your arrow keys do not work, run `term windows` for this session and see Other hardening. An account created before the `WINDOWS` definition shipped keeps its old setting until `update.accounts`.

`term` also reports the page size, and SD's default is `120 × 36` — not `80 × 24`. The shipped dictionaries and the default `list` layouts are formatted for 120 columns, so **a console window narrower than that makes ordinary reports look wrapped or truncated** and the report is not at fault. Widen the window, or set it for the session with `term default` — which puts the `120 × 36` back and prints nothing while doing it, so check with a bare `term` afterwards. `term 120,36` is the same thing typed out. See Other hardening.

4. Make a file and put something in it

```
create.file customers
ed customers 1001
```

`ed` is the **line** editor, and it needs nothing installed. If you would rather have a full screen, `edit` opens the same record in Microsoft Edit and `micro` opens it in micro — see Programmer commands. The old full-screen editors `sed`, `update.record` and `modify` are all gone; see Not in SD Core.

In `ed`: `i` to insert, type your lines, a full stop on its own line to stop inserting, then `fi` to file and exit.

You do not have to write programs in `ed`. An account's `bp` file is a **directory file** — an ordinary Windows folder with one file per program — so Notepad++, VS Code or any text editor works on it just as well:

```
C:\ProgramData\SD\user_accounts\<account>\bp
```

Save the file, then `basic` and `catalog` it from inside SD as usual. SD folds CR+LF line endings on the way in, so a Windows editor's output needs nothing done to it — see Other hardening.

```
list customers
count customers
```

Commands are lower case now, and so are the VOC records behind them. Typing `LIST` still works — SD tries what you typed, then lower case, then upper. See Lower case.

THIS IS THE POINT AT WHICH MOST THINGS SHOULD FEEL LIKE OpenQM. If anything in ordinary data work behaves differently and is not described in this set, that is worth reporting.

5. Become an administrator

```
logto sdsys
```

You will get a UAC consent prompt unless the session is already elevated. That is the gate — there is no SDSYS password, and there is deliberately no second shared secret held by every administrator.

IF YOU ARE OVER ssh, THIS MAY NOT WORK. A UAC prompt has no interactive desktop there. Start an elevated terminal at the machine instead. See Security.

6. Create an account for somebody else

```
create.account user jane programmer both
```

Three things about that line, and each has caught people out:

`programmer`

without it, `jane` gets a **standard** account and cannot compile, catalogue, edit or create files. See Account types

`both`

one of `ssh`, `api`, `both`, `none` is **required**. There is no default

it needs elevation

creating a Windows account cannot be done from a filtered token

You will be prompted for Jane's password, masked. **Refusing the prompt creates nothing at all** — a user account cannot exist without a password.

The password is for the API. Console and ssh logins ask for nothing.

7. Sign in as Jane

```
ssh jane@localhost
```

You land directly inside SD, not at a Windows prompt. That is the forced command, and it applies to everyone who connects, administrators included.

Jane cannot log in to Windows at this machine. She is denied the console and Remote Desktop by group membership, deliberately. She reaches SD over ssh or through an API client — which is what the `ssh` / `api` / `both` keyword chose. See ssh access.

`ssh localhost` needs no network and works on a machine with no network connection at all.

8. Leave

```
off
```

What to try next, in rough order of how likely it is to find something

1. **Your own application data. There is no restore utility**, so the way in is a short BASIC program that reads your exported data and writes the records. Then query it — the query processor is where most of the surface area is.
 2. **A client program against the API.** Point it at port 4243 — **not** an ssh tunnel any more. It needs a client library from this release, because the old cleartext login is gone. **The architecture must match the application**, and **The DLL goes in the same directory as the application that loads it** — Windows searches there before `PATH`. **If you have no client of your own, mvDeveloper is free** and is a 32-bit application, so it wants the 32-bit `qmclilib.dll`: <https://www.brianleach.co.uk/mvDeveloper>. See API access and Client distribution.
 3. **A standard account.** Create one without `programmer` and confirm it can run your application but not edit it.
 4. **An upgrade.** Install over the top and check your data survived and `update.accounts` brought the VOC forward.
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When something goes wrong

Something SD did, and who did it

`audit`, in `C:\ProgramData\SD\sdsys`

Diagnostics, and API connections

`errlog`, same place

"the account was not created — what happened"

`sd-elevate.log`, in `C:\ProgramData\SD`

The installation itself

Check the SD installation on the Start Menu

Other hardening explains which log answers which question — they are not interchangeable.

When you report something, say which build. The release stamp is on the sign-on banner and in `C:\Program Files\SD\changelog`.